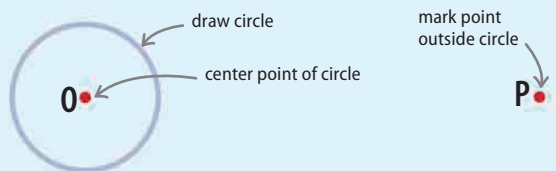
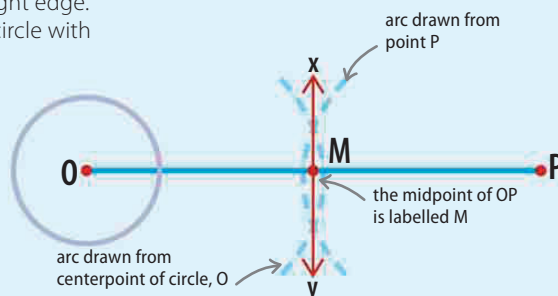


Constructing tangents

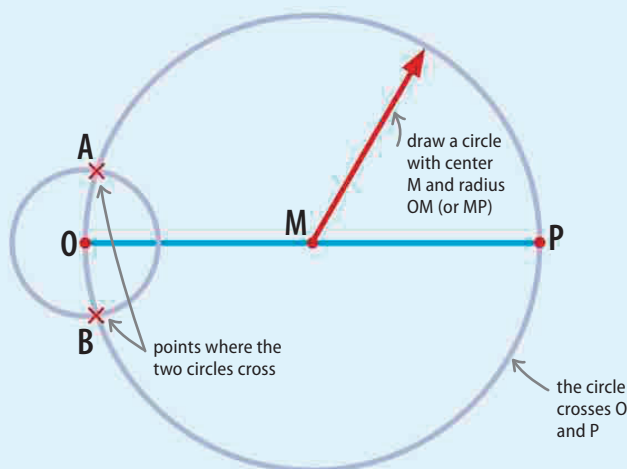
To construct a tangent accurately requires a compass and a straight edge. This example shows how to construct two tangents between a circle with center O and a given point outside the circle, in this case P .



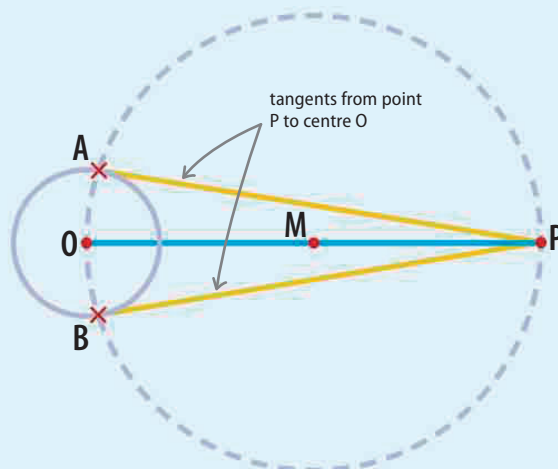
Draw a circle using a compass, and mark the center O . Also, mark another point outside the circle and label it (in this case P). Construct two tangents to the circle from the point.



Draw a line between O and P , then find its midpoint. Set a compass to just over half OP , and draw two arcs, one from O and one from P . Join the two points where the arcs cross with a straight line (xy). The midpoint is where xy crosses OP .



Set the compass to distance OM (or MP which is the same length), and draw a circle with M as its center. Mark the two points where this new circle intersects (crosses) the circumference of the original circle as A and B .



Finally, join each point where the circles intersect (cross), A and B , with point P . These two lines are the tangents from point P to the circle with center O . The two tangents are equal lengths.

Tangents and angles

Tangents to circles have some special angle properties. If a tangent touches a circle at B , and a chord, BC , is drawn across the circle from B , an angle is formed between the tangent and the chord at B . If lines (BD and CD) are drawn to the circumference from the ends of the chord, they create an angle at D that is equal to angle B .

▷ Tangents and chords

The angle formed between the tangent and the chord is equal to the angle formed at the circumference if two lines are drawn from either end of the chord to meet at a point on the circumference.

